

Angular impulse and angular momentum

The equation of rotation of rigid body is $I\ddot{\theta} = \Sigma M$

$$I \frac{d\dot{\theta}}{dt} = \Sigma M \gggg I d\dot{\theta} = \Sigma M dt \quad \text{or} \quad \frac{d(I\dot{\theta})}{dt} = \Sigma M \gggg \frac{dH}{dt} = \Sigma M$$

Integrating both sides

$$\int_{\dot{\theta}_0}^{\dot{\theta}} I d\dot{\theta} = \int_0^t \Sigma M dt$$

$$I\dot{\theta} - I\dot{\theta}_0 = \int_0^t \Sigma M dt$$

This shows that the change in angular momentum of a body is equal to the angular impulse of the external moment ΣM

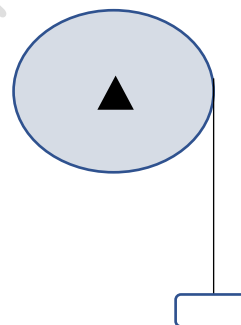
$$I\dot{\theta}_0 + \int_0^t \Sigma M dt = I\dot{\theta}$$

Initial angular momentum + angular impulse = final angular momentum.

If the angular impulse of the moment is zero (or $\Sigma M = 0$), then the angular momentum remains constant. It is known as the principle of conservation of angular momentum

Example 1

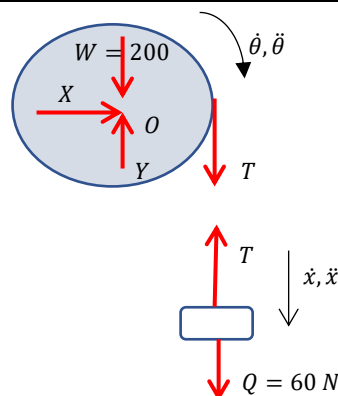
A block of weight 60 N is attached to a cord which is wrapped around the periphery of a disc of weight 200 N and radius 0.2 m. If the block is initially moving down with a speed of 2 m/s, determine its speed in 3 sec.



The free-body diagrams of the disc and the block are shown. The block is given an initial velocity $v_0 = 2 \text{ m/s}$ down. Therefore, the initial angular velocity of the disc, $\omega_0 = \frac{v_0}{r} = 10 \frac{\text{rad}}{\text{s}}$ clockwise

Let v be the velocity of the block after 3 s and ω be the corresponding angular velocity of the disc. By the angular impulse-momentum principle applied to the disc,

$$I_0 \dot{\theta}_0 + \int_0^t \Sigma M_O dt = I_0 \dot{\theta}$$



$$I_0 = \frac{W r^2}{g} = 0.4 \text{ kg m}^2$$

$$\Sigma M_O = T \times r = 0.2T$$

$$I_0 \dot{\theta}_0 + \int_0^t \Sigma M_O dt = I_0 \dot{\theta} \gggg 0.4 \times 10 + \int_0^3 0.2T dt = 0.4\dot{\theta}$$

$$4 + 0.6T = 0.4 \frac{v}{0.2} \gggg 4 + 0.6T = 2v$$

Applying the linear impulse-momentum principle to the block, we can write

$$\frac{Q}{g} v_o + \int_0^t \Sigma F dt = \frac{Q}{g} v$$

$$\frac{60}{9.81} \times 2 + \int_0^3 (60 - T) dt = \frac{60}{9.81} v$$

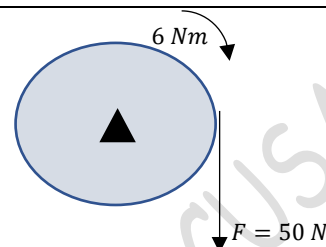
$$12.2 + (180 - 3T) = 6.11 v \gggg 192.2 - 3T = 6.11v$$

Solving the two equations, we get $v = 13.2 \text{ m/s}$

#Exercise 43

43.1

A disc of weight 100 N and radius $r = 0.25 \text{ m}$ supported at its centre is acted up on by a constant moment 6 Nm and a force of 50 N is applied to a cord wrapped around its periphery. Determine its angular velocity after 5 s starting from rest.



43.2

A wood block of weight $W = 22 \text{ N}$ rests on a horizontal plane and is attached to a fixed hub O by a horizontal string as shown. A revolver bullet of weight $w = 0.28 \text{ N}$ is fired horizontally in to the centre of the block in the direction normal to the plane of the figure with velocity $v = 600 \text{ m/s}$. Calculate the number of revolutions the block will make around the vertical z -axis before coming to rest if the coefficient of friction between the block and the plane is 0.25 . Neglect the dimensions of the block



Prof. Biju N, School of Engineering, CUSAT